Research Assignment 02

Data science foundations, data types, EDA, preprocessing, and Titanic dataset analysis

Course	**Hope to Skills — Free Artificial Intelligence Advance Course**
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Academic submission prepared in a structured report format

Document field	Details
Research assignment	Research Assignment 02
Student name	Abdul Rehman
Course	Hope to Skills — Free Artificial Intelligence Advance Course
Instructors	Irfan Malik and Dr. Sheraz
Objective	To explain the fundamentals of data science, data types, exploratory data analysis, data preprocessing, and an initial study of the Titanic dataset used in the course.

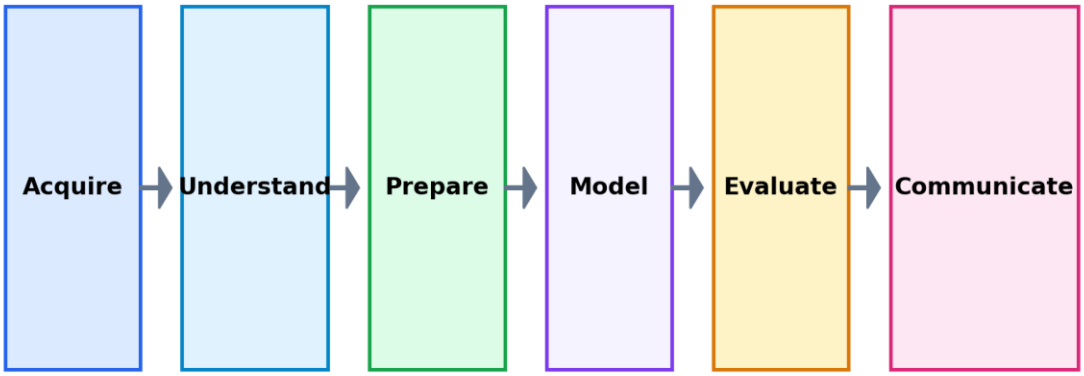
Report overview

This submission is written in a structured academic style and answers every question in the assignment directly.
I used concise explanations first, then added practical reasoning, advantages, limitations, and application context so the report shows both conceptual understanding and applied thinking.

Question 1 — Define Data Science and explain its important points

Data Science is the discipline of extracting useful knowledge, patterns, and actionable insights from data by combining statistics, computing, and domain understanding. In simple terms, it is the process of turning raw data into evidence that can support decisions, predictions, and intelligent systems.

Data science is not only about building models. It includes the full pipeline: collecting data, understanding it, cleaning it, exploring it, selecting suitable methods, evaluating results, and finally communicating the findings to decision-makers in a form they can actually use.



Data science is an end-to-end process that converts raw data into decisions and action.

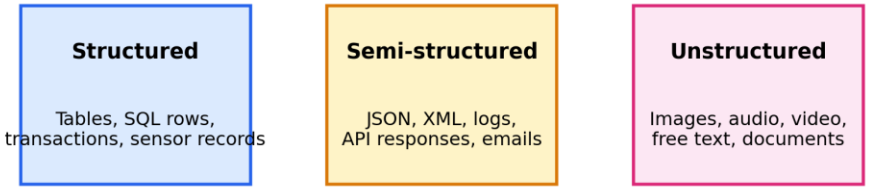
Important key point	Explanation
Interdisciplinary nature	Data science combines statistics, programming, machine learning, and domain knowledge.
Data-driven decision making	It helps organizations replace guesswork with measurable evidence.
Problem-solving orientation	The purpose is not only to analyze data but to answer a real business or

	research question.
Communication and storytelling	Insights are only valuable if they can be communicated clearly to non-technical stakeholders.
Iterative workflow	Data science is rarely one straight path; analysts often revisit earlier steps after new findings appear.

In practice, this means a data scientist must think like an analyst, engineer, communicator, and problem solver at the same time.

Question 2 — Different types of data with real-world examples

Data is commonly grouped into structured, semi-structured, and unstructured forms. The difference is mainly based on how clearly the data is organized and how easy it is to store, query, and analyze with traditional tools.



The main difference is how clearly the data is organized and how much preprocessing is needed before analysis.

Data type	Description	Real-world examples	Analysis style
Structured	Highly organized into rows and columns with a fixed schema	Bank transactions, sales records, student marks, inventory tables	SQL queries, dashboards, classical ML, tabular statistics
Semi-structured	Has some organization but not a strict table schema	JSON API responses, XML files, web logs, emails	Parsing plus transformation before analysis
Unstructured	Does not follow a fixed tabular format	Images, audio, video, social media text, PDFs	NLP, computer vision, embeddings, deep learning

Structured data is easiest to analyze because each attribute is already organized. Semi-structured data needs parsing but still contains useful tags or keys. Unstructured data usually needs the most preprocessing because meaning must be extracted from raw content such as text, images, or sound.

Question 3 — Importance of Exploratory Data Analysis (EDA)

EDA is the process of examining data before model training in order to understand its shape, quality, patterns, and possible problems. It is important because poor understanding at the start usually leads to poor modeling decisions later. EDA helps answer practical questions such as: Are values missing? Are there outliers? Is the target balanced? Which variables appear useful?

Three important ideas inside EDA are central tendency, dispersion, and proximity.

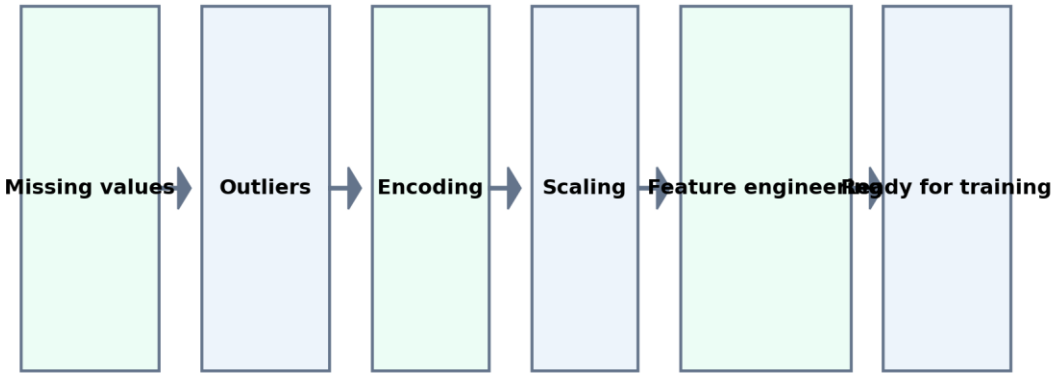
EDA concept	Definition	Examples and use

Central tendency	Measures the center of the data	Mean, median, and mode help summarize the typical value of a variable.
Dispersion	Measures the spread or variability of the data	Range, variance, standard deviation, and IQR show whether values are tightly grouped or widely scattered.
Proximity	Measures similarity or distance between observations	Euclidean distance, cosine similarity, and correlation help compare records or variables.

Why EDA matters: If central tendency is very different from the median, the distribution may be skewed. If dispersion is too high, outliers may be present. If variables are strongly correlated, they may contain overlapping information. These early findings directly influence preprocessing and modeling choices.

Question 4 — Core points of Data Pre-Processing and why it is necessary

Data preprocessing is the step where raw data is cleaned, transformed, and prepared so that a machine learning model can use it effectively. Real-world datasets are rarely perfect; they often contain missing values, inconsistent formats, duplicated rows, noisy inputs, outliers, or categorical features that must be encoded.



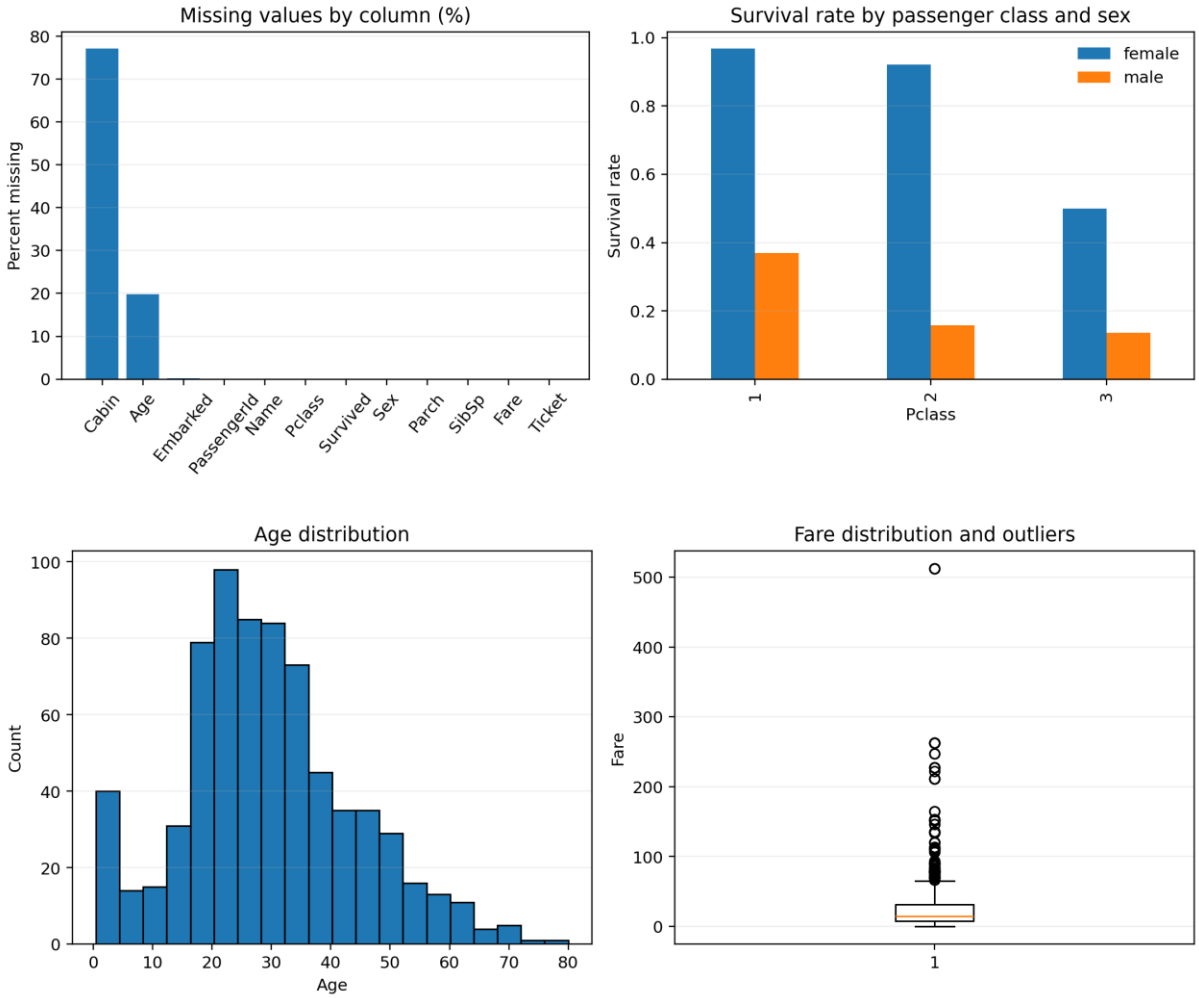
Preprocessing improves data quality, consistency, and model readiness before training begins.

Core preprocessing point	Why it matters
Handling missing values	Prevents models from failing or learning biased patterns from incomplete data.
Outlier treatment	Stops extreme values from distorting averages, distances, or model coefficients.
Encoding categorical variables	Converts labels such as male/female or embarked port into numeric form that models can use.
Scaling and normalization	Ensures features with large numeric ranges do not dominate distance-based or gradient-based models.
Data cleaning and consistency checks	Fixes invalid values, duplicate records, and mismatched formats.
Feature engineering	Creates stronger predictive variables from the raw attributes.

Preprocessing is necessary because model quality depends heavily on input quality. A strong algorithm trained on messy data will still give weak results, while a well-prepared dataset often allows even simple models to perform very well.

Question 5 — Initial study of the Titanic dataset

The Titanic dataset used in the course contains 891 rows and 12 columns. It is a classic binary classification dataset where the goal is to predict whether a passenger survived or not. The target variable is Survived, and because its values are 0 and 1, the problem is clearly a classification problem, not a regression problem.



My first observation is that the dataset is not fully cleaned. The Cabin column is missing in about 77.1% of rows, Age is missing in about 19.9% of rows, and Embarked has a small number of missing entries. This means preprocessing is required before reliable model training can begin.

Important attribute	Why it is important
Pclass	Passenger class is strongly linked to survival because it reflects travel conditions and access to evacuation.
Sex	This is one of the strongest predictors; in the dataset, female passengers have much higher survival rates than male passengers.
Age	Age matters because children and older passengers may have faced different survival probabilities and support needs.
Fare	Fare can act as an indirect indicator of socioeconomic status and class-related access.

Embarked	Port of embarkation may capture group-wise and demographic patterns.
SibSp and Parch	These help estimate family structure, companionship, and whether a passenger was traveling alone.
Less important raw attribute	**Reason**
PassengerId	This is mainly a unique identifier and usually has no predictive meaning.
Ticket	The raw ticket string is difficult to use directly without heavy feature engineering.
Cabin	The column may contain useful information, but it has too many missing values in raw form.
Name	The full raw name is not directly model-ready, although titles such as Mr, Mrs, or Miss could be extracted from it.

Initial observations about the data: The survival classes are not perfectly balanced because 549 passengers did not survive while 342 survived. Survival rates differ strongly by sex and passenger class, which suggests that those variables will likely be highly informative. The fare variable also contains clear outliers, and age contains missing values that must be handled carefully.

EDA steps needed to make the dataset cleaner:

- Handle missing values in Age, Cabin, and Embarked using an appropriate strategy such as imputation, indicator variables, or dropping highly incomplete columns.
- Check for outliers in Fare and potentially in Age, then decide whether clipping or transformation is needed.
- Encode categorical columns such as Sex and Embarked into numerical form.
- Consider feature engineering such as family size, title extraction from Name, or an is-alone flag.
- Study class balance and evaluate the train/test split carefully so the target distribution remains reasonable.

Titanic dataset conclusion

The Titanic dataset is suitable for an introductory classification problem because it contains a clear binary target, meaningful tabular features, and realistic preprocessing challenges such as missing values, categorical encoding, and feature relevance.

It is not fully cleaned in raw form, so EDA and preprocessing are necessary before model training. That is exactly why it is useful in teaching the data science workflow.